

# Math 104– Lecture 01

Keshav Ramamurthy

September 2, 2026

## 1 START

**Definition 1.1** (Sequence). A sequence is a map from  $n \in \mathbb{Z} | n \geq m (m \in \mathbb{Z})$  to a set  $X$

**Definition 1.2** (Convergence). A sequence  $(s_n)_1^\infty \subset \mathbb{R}$  converges to  $s \in \mathbb{R}$  if

$$\forall \epsilon > 0, \exists N > 0 \text{ such that } n > N \implies |s_n - s| < \epsilon$$

**Definition 1.3** (Divergence). A sequence  $(s_n)$  diverges if  $(s_n)$  doesn't converge to any real number.

**Definition 1.4** (Limit Notation). We write

$$\lim_{n \rightarrow \infty} (s_n) = s, \lim s_n = s, s_n^{n \rightarrow \infty} \rightarrow s$$

We write

$$\lim_{n \rightarrow \infty} (s_n) = \infty$$

if

$$\forall m \in \mathbb{R} \exists N > 0 \text{ such that } n > N \implies s_n > m$$

**Example 1.5** (Convergence Example). Take  $s_n = \frac{3n+1}{7n+4}, n \in \mathbb{N}$ . We have

$$\begin{aligned} \lim_{n \rightarrow \infty} (s_n) &= \lim_{n \rightarrow \infty} \frac{3n+1}{7n+4} \\ &= \lim_{n \rightarrow \infty} \frac{3 + \frac{1}{n}}{7 + \frac{4}{n}} \\ &= \frac{3}{7} \end{aligned}$$

How do we verify this?

*Proof.* Epsilon Delta: Let  $\epsilon > 0$ . Take  $N = \frac{19}{49\epsilon} + \frac{4}{7}$ . if  $n > N \rightarrow |s_n - \frac{3}{7}| < \epsilon$

The real step is finding  $N$  with equation

$$\left| \frac{3n+1}{7n+4} - \frac{3}{7} \right| < \epsilon$$

□

## 2 Main idea

*Write the one-sentence point of today's lecture here.*

## 3 Definitions and notation

**Definition 3.1** (First definition). Write the definition and one sentence explaining why it matters.

## 4 Claims, theorems, and proof sketches

**Claim.** State the claim in one precise sentence.

**Proof sketch.** Record the key idea, the first useful equation, and the step that still needs checking.

## 5 Examples and calculations

**Example 5.1** (Lecture example). Write the setup, the calculation, and the conclusion.

## 6 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: